

BASIC INTRODUCTION (1-2)

Transportation networks formed through positive feedback are ubiquitous across a range of ecological and sociological domains.

An ant leaves behind a pheromone trail, a bear forges a way through undergrowth, or a person tramples down grass; in these cases and many others, where one individual travels, others are more likely to follow.

MORE DETAILED BACKGROUND (2-3)

Active-walker models simulate this process and show that individuals following simple behavioral rules produce trail systems with many of the features of real-world networks.

[Another sentence here about ant simulations & theory?]

GENERAL PROBLEM (1)

However, agents in existing models act on local information only, and little has been done to investigate walkers with more complete knowledge of their environment and the abilities to plan their behavior intelligently.

HERE WE SHOW (1)

Here we investigate a model in which walkers navigate a fully-revealed environment and take a least-cost path to their destination.

MAIN RESULT (2-3)

We show that such walkers create intricate networks across a range of scenarios. When presented with different model parameters, walkers produce efficient networks that minimize travel costs for a given amount of sustainable trail. We show that a simple relationship exists between the relative cost of travelling on and off the trails and the internal geometries of the networks. And we demonstrate that developing trail networks may exhibit one or multiple attracting states.

GENERAL CONTEXT (1-2)

Our results illuminate key features of networks likely to be produced by higher-order animals, and provide a theoretical framework useful in understanding behavior of migratory animals, designing urban spaces, and even illuminating some features of cities.